

Cassie Liang

AI Application Engineer (Agents / RAG / LLM Systems)

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EDUCATION

Northeastern University | M.S. in Artificial Intelligence · GPA 3.75/4.0 | Graduate TA, Generative AI Expected Dec 2027
Shanghai University of Finance and Economics | B.S. in Accounting | Huawei & People's Scholarships Jun 2023

TECHNICAL SKILLS

LLM / Agents: RAG, hybrid retrieval (pgvector dense + Postgres full-text sparse + RRF), MCP, structured outputs, MLX LoRA, evaluation
Backend / Data: Python, FastAPI, PostgreSQL / pgvector, Redis, Celery (at-least-once delivery, idempotent consumers), Docker, SQL, p
Multimodal / Client: CLIP, ASR, VLM, TTS, React, Swift / SwiftUI, AVFoundation, Chrome Extension MV3

PROJECTS

MindBridge | Local-first long-term memory engine for AI agents · Independent [GitHub](#) | [Demo](#)

Python · FastAPI · MCP · PostgreSQL / pgvector · Qwen2.5-3B · MLX LoRA · Redis

- Reliable ingestion:** Parse Claude Code / Codex CLI logs by byte cursor; ingested **13,072 turns / 242 session cards**.
- Memory model:** Split T1 turns, T2 cards, and T3 temporal preferences; cursor + source_key make rebuilds idempotent.
- Unified service:** Reuse MemoryService across MCP and HTTP to prevent Agent-tool and server-side logic drift.
- Local tuning:** Trained MLX LoRA on 198 rows; 45-case date holdout reached **86.7%** first-pass schema validity vs. teacher **82.2%**.

Vance | AI fitness companion with executable training plans · Independent [GitHub](#) | [Demo](#)

SwiftUI · SwiftData · Cloudflare Workers / D1 · Kimi K2.6 · SSE · MiniMax TTS

- Agent loop:** Execute weight updates, exercise swaps, and plans via tool_use / tool_result; return outcomes to the model.
- Constrained planning:** Validate against **1,324** D1 seeds and **50** curated on-device exercises; filter unsafe options.
- Service architecture:** Workers protect model keys and stream typed SSE; iOS runs only allowlisted actions.
- Multimodal UX:** Combine speech input, MiniMax TTS, and Kimi equipment vision; admit only high-confidence context.

CorpCheck | SEC filing retrieval with deterministic abstention · Individual rebuild [GitHub](#) | [Landing](#)

Python · FastAPI · PostgreSQL / pgvector · Postgres FTS (ts_rank) · RRF · MCP · FinanceBench

- Retrieval scale:** Fuse pgvector dense search with Postgres full-text ranking (ts_rank) via RRF across **1,662 filings / 469,874 chunk**
- Version governance:** Apply section-aware amendment filtering before ranking and citation.
- Deterministic abstention:** Before the LLM, score top-1 / top-3 similarity and evidence coverage; return explicit refusals.
- Evaluation:** Cut FinanceBench queries with no same-filing top-10 evidence from 19/35 to 4/35.

1Day | Local-first collaborative video diary for iOS · Independent · App Store v1.1 [GitHub](#) | [App Store](#)

Swift · SwiftUI · AVFoundation · CloudKit · Sign in with Apple · XCTest

- Product scope:** Support **1-day / 7-day** journeys and **2 / 5 / 10-second** clips to bound capture effort.
- On-device rendering:** Normalize orientation and tracks; compose audio, transitions, captions, and MP4.
- Collaboration consistency:** Derive CloudKit IDs from room + participant + slot; retakes update the same record.
- Engineering proof:** Passed **28 / 28** XCTest cases across state, migration, and real-fixture MP4 export.

EXPERIENCE

Smith-Kettlewell Eye Research Institute | AI Product & Engineering Intern San Jose | Jan 2025 - Present

React · Python · FastAPI · Whisper · VLM / LLM · TTS

- AI workflow:** Built scene-level React review and FastAPI generation flows spanning Whisper, VLM / LLM, and TTS.
- Measured outcome:** Analyzed a 30-user study against the baseline editor; manual authoring time fell **64%**.

Deloitte | Audit Assistant Shanghai | Jun 2023 - Dec 2024

Python · VBA · Excel Automation · SQL

- Automation:** Built Python workbook mapping for **30,000+** ledger rows, improving engagement review efficiency by **60%**.
- Reuse:** Packaged 20+ controls into configurable workflows covering **90%** of common errors across 10+ clients.

Award project | VisoCode: LangGraph agents and Docker sandboxes generate Manim videos; AdventureX 2025 Best Technical Award. [GitHub](#)
Becoming: Shipped an AI growth-coach MVP in 48 hours across product and full stack; Startup Forest Second Prize.

Other | Cassie Capture: Streams Content Script → Service Worker → Offscreen to split extra-long webpages into stable PDFs. [GitHub](#)
Tabspace: Local-first Chrome new-tab workspace for tab groups, bookmarks, and review-before-apply. [GitHub](#)